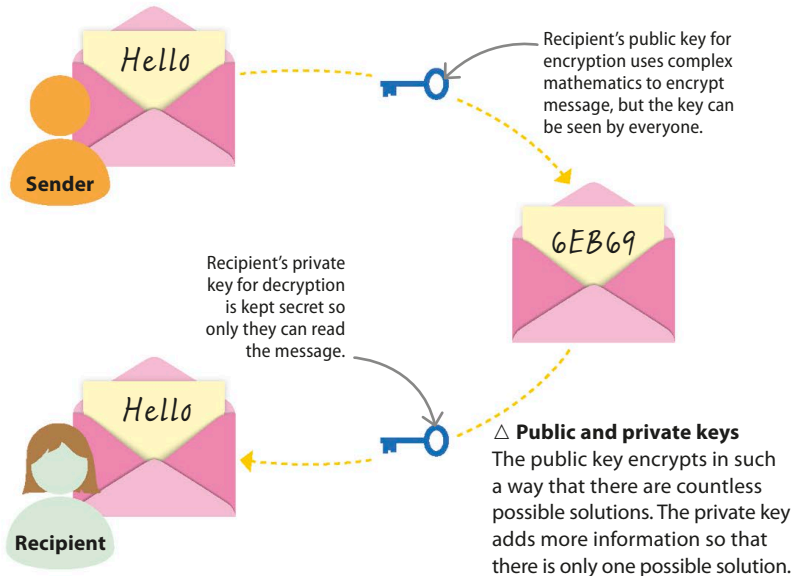


Public-key encryption

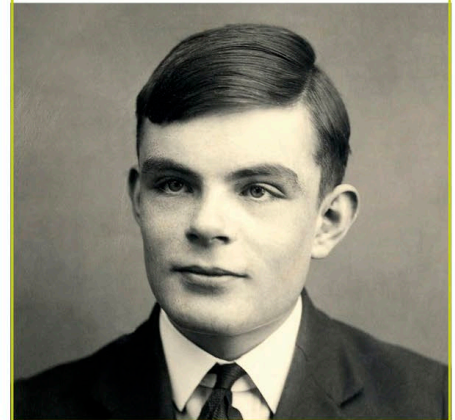
The problem with early forms of encryption was that they could be easily intercepted and decrypted. Public-key encryption was developed in the 1970s and avoids this. Essentially, both parties have two keys – a public one that is used to encrypt a message, and a private one that is known only by the sender and the recipient.



BIOGRAPHY

Alan Turing

English computer scientist Alan Turing (1912–1954) led the team that broke the Enigma code used by the German navy during WWII. He also developed the Turing Test, where a computer is considered intelligent if, during a blind test, an evaluator cannot tell if it is a human or an artificially intelligent computer/machine.



LINGO

Useful terms

Cryptography: The study of creating and breaking secret codes.

Cipher: An algorithm used to encrypt or decrypt a message.

Key: Extra information used with ciphers to encrypt or decrypt a message.

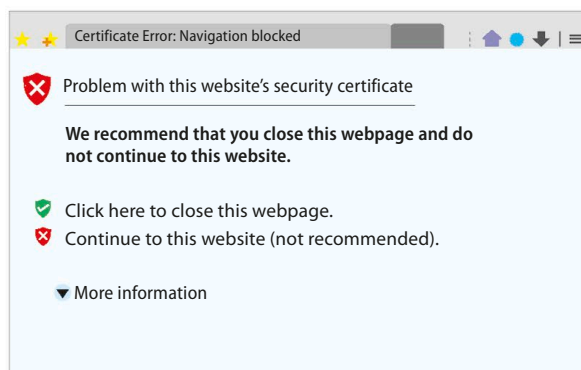
Decryption: The process of using a cipher and key to reveal the meaning of an encrypted text.

Plaintext: An unencrypted piece of text.

Ciphertext: An encrypted piece of text.

Secure Socket Layer

Encryption is added to internet connections by adding a Secure Socket Layer (SSL) to the normal communication procedures. The URL of a website using SSL starts with https instead of http. Any website that wants to allow users to connect to it securely needs an SSL Certificate from a recognized provider. This includes a public and private key pair, which allows the site to encrypt the traffic between it and its users. Many email applications also use SSL to ensure users' emails are secure as they travel across the web.



SSL Certificate

If a site's SSL Certificate is outdated or unrecognized, web browsers will display a warning. Some browsers will also prevent users from viewing the page.